

Notes For MLA Cryptography Workshop

Kuan Yue Han 25J13

Zhang Hengwei 25J17

Mathematical Cryptography

1. Basic Ciphers

1.1. Caesar cipher

Let the set of letters be $S = \{A, B, C, \dots, Z\}$. This cipher uses a bijective function $\sigma : [0, 25] \rightarrow [0, 25]$

$c \mapsto c + k \pmod{26}$ for some constant key k

For example, set $k = 2$, the word

“BRANSON” is $S_1 S_{17} S_0 S_{18} S_{14} S_3 \rightarrow S_{\sigma(1)} S_{\sigma(17)} S_{\sigma(0)} S_{\sigma(18)} S_{\sigma(14)} S_{\sigma(3)} \rightarrow$ “EUDQVRQ”. However, this cipher is easily broken when the trick is known as there are only 26 possible keys (k).

1.2. Worksheet Example (Caesar)

Task: Encrypt the plaintext “MEET AT NOON” using key $k = 7$.

Letter $\leftrightarrow$ number guide (A=0, B=1, ..., Z=25):

Letter:	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Digit:	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Cipher:	H	I	J	K																						

Fill in the table for MEET AT NOON:

Plaintext:	M	E	E	T	—	A	T	—	N	O	O	N
Digit:												
Ciphertext:												

1.3. Affine Cipher

Let the set of letters be $S = \{A, B, C, \dots, Z\}$. This cipher uses a bijective function $\sigma : [0, 25] \rightarrow [0, 25]$

$c \mapsto ac + k \pmod{26}$ for some constants a and k where $\gcd(a, 26) = 1$. (this ensures bijectivity)

For example set $a = 5$ and $k = 8$, the word

“WACKY” is $S_{22} S_0 S_2 S_{10} S_{24} \rightarrow S_{\sigma(22)} S_{\sigma(0)} S_{\sigma(2)} S_{\sigma(10)} S_{\sigma(24)} \rightarrow$ “VGNKB”.

This cipher is slightly more secure than the Caesar cipher as there are more possible keys (a, k) but it is still easily broken when the trick is known as there are only 12 possible values for a and 27 possible values for k , giving a total of 324 possible keys.

1.4. Worksheet Example (Affine)

Use the same parameters as the example above: $a = 5$, $k = 8$.

Fill in the affine mapping table ($A=0, \dots, Z=25$) with modulo 26:

Letter:	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
x:	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
$5x + 8$:	8	13	18	23	2	7																				

Now do a word: Encrypt the plaintext “MATH” using the same a, k .

M	A	T	H

Ciphertext:

2. Group Properties

A **group** is a set G together with a binary operation $*$ such that:

1. **(Closure)** For all $a, b \in G$, the result $a * b \in G$.
2. **(Associativity)** For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.
3. **(Identity element)** There exists an element $e \in G$ such that $a * e = e * a = a$ for all $a \in G$.
4. **(Inverse element)** For every $a \in G$, there exists an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

An important example in cryptography is the set of permutations of a finite set. The set of all permutations on n elements forms a group under composition, called the symmetric group S_n .

These permutations can be composed and inverted, meaning that encryption and decryption operations form a group under composition. This algebraic structure helps explain why certain ciphers are easy or difficult to break.

2.1. Cyclic group and generators

A group G is called cyclic if there exists an element $g \in G$ such that every element of G can be expressed as a power of g . In other words, $G = \{g^k \mid k \in \mathbb{Z}\}$. The element g is called a generator of the group.

$$G = \{g^1, g^2, g^3, \dots, g^n\}$$

In the group $\mathbb{Z}/p\mathbb{Z}$, there are exactly $\varphi(p-1)$ **generators**, where φ is the Euler totient function. We say that $\mathbb{Z}/p\mathbb{Z}$ is a cyclic group because it can be “generated” by a single element.

3. Extended Euclidean Algorithm (and the Euclidean Algorithm)

The Euclidean Algorithm is used to find the greatest common divisor of two numbers. Notice that the greatest common divisor of two numbers is one when the numbers do not share any common factors other than one (hence are **coprime**). The converse is also true, hence we say that two numbers are **coprime** if and only if their greatest common divisor is one.

First notice that $\gcd(a, b)$ is the greatest common divisor of a and b . Let

$$g = \gcd(a, b)$$

Then we can write $a = ga'$ and $b = gb'$ for some integers a' and b' . If $g \mid a$ and $g \mid b$, then $g \mid a - kb$ for any integer k . In other words,

$$\begin{aligned}\gcd(a, b) &= \gcd(b \bmod a, a) \\ &= \gcd(b_0, a) \\ &= \gcd(a \bmod b_0, b_0) \\ &= \gcd(a_0, b_0) \\ &= \dots\end{aligned}$$

when $b \bmod a$ is taken to be the remainder when b is divided by a .

The algorithm will eventually terminate when it reaches $b_k = 0$ or $a_k = 0$ for some integer k , in which case, $g = a_k$ or $g = b_k$ respectively.

First we illustrate the Extended Euclidean Algorithm and the original Euclidean Algorithm. set $a = 73$, $b = 25$.

$$\begin{aligned}73 &= 25 \times 2 + 23 \\ 25 &= 23 \times 1 + 2 \\ 23 &= 2 \times 11 + 1 \\ 2 &= 1 \times 2 + 0\end{aligned}$$

Thus, the greatest common divisor of 73 and 25 is 1. Notice how each step can be represented as a linear combination of the previous step.

$$23 = a - b \times 2 \quad (1)$$

$$2 = b - 23 \times 1 \quad (2)$$

$$= b - (a - b \times 2) \times 1$$

$$1 = 23 - 2 \times 11 \quad (3)$$

$$= 23 - (b - (a - b \times 2) \times 1) \times 11$$

$$= a - b \times 2 - (b - (a - b \times 2) \times 1) \times 11$$

The gcd is just a linear combination of a and b .

It can be shown that it takes at most $2 \log(a) + 1$ steps to terminate, and the average number of steps is approximately

$$\frac{12 \ln(2)}{\pi^2} \ln(a).$$

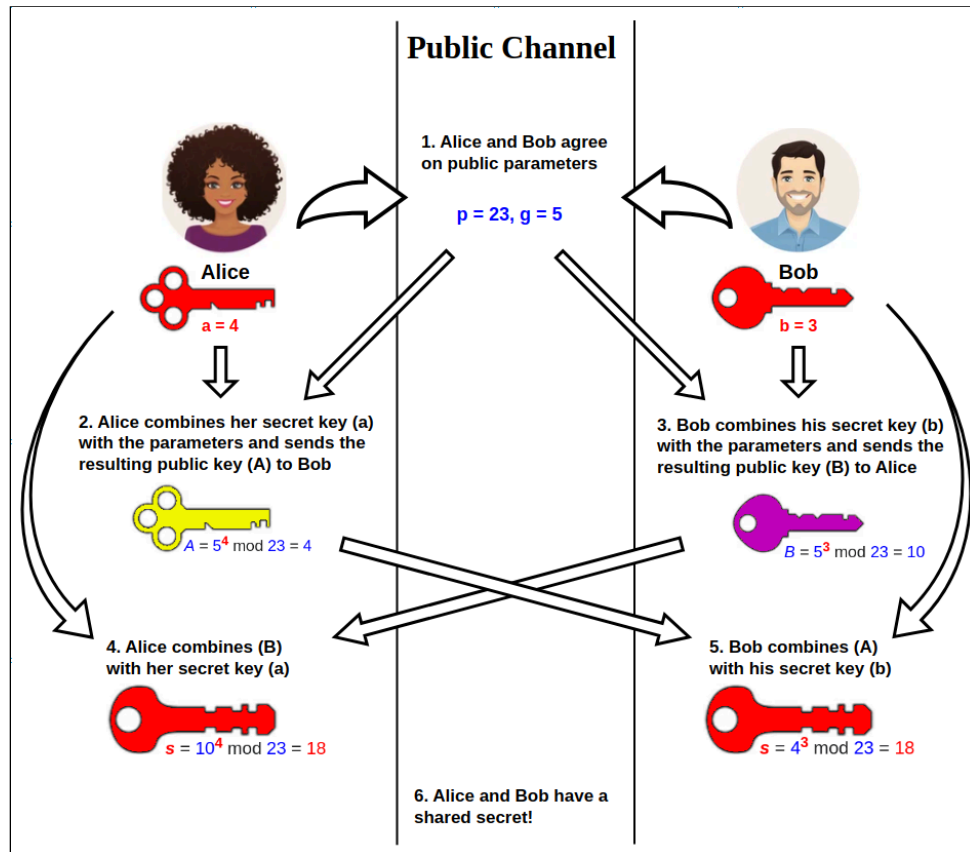
4. Discrete Logarithm Problem(DLP)

The DLP is the problem of finding an integer k such that $g^k \equiv h \pmod{p}$, given g , h , and a prime p . If one can find such a solution, then there must be infinitely many solutions. As given in the FLT, assume k is a solution

$$\begin{aligned}g^k &\equiv h \pmod{p} \\ g^{k+m(p-1)} &\equiv g^k g^{m(p-1)} \equiv g^k \equiv h \pmod{p}\end{aligned}$$

Thus, the Discrete Logarithm Problem is only defined modulo $p - 1$. It is widely accepted that the DLP is hard to solve. Classical algorithms like the Baby-step Giant-step algorithm and Pollard's rho algorithm still take roughly $O(\sqrt{p})$ time to solve the DLP.

5. Diffie-Hellman



Diffie-Hellman(DH) protocol allows two parties (Alice and Bob) to establish a shared secret over an insecure channel. The security of the protocol relies on the difficulty of the discrete logarithm problem (DLP).

Secret values are **bolded** for better visualisation.

Setup

- Prime modulus: $p = 23$
- Generator: $g = 5$

This generator has to generate a large subgroup of the multiplicative group of integers modulo p to ensure security. In this case, 5 is a primitive root modulo 23, meaning it generates the entire group of nonzero elements modulo 23.

5.1. Alice

Alice chooses a secret:

$$\mathbf{a} = 4$$

She computes:

$$A \equiv g^a \pmod{p}$$

$$A \equiv 5^4 \equiv 4 \pmod{23}$$

She sends A to Bob.

5.2. Bob

Bob chooses a secret:

$$b = 3$$

He computes:

$$B \equiv g^b \pmod{p}$$

$$B \equiv 5^3 \equiv 10 \pmod{23}$$

He sends B to Alice.

5.3. Shared Secret Calculation

Alice computes:

$$s \equiv B^a \pmod{p}$$

$$s \equiv 10^4 \equiv 18 \pmod{23}$$

Bob computes:

$$s \equiv A^b \pmod{p}$$

$$s \equiv 4^3 \equiv 18 \pmod{23}$$

5.4. Result

Both obtain the same shared secret:

$$s = 18$$

It can be shown that the shared secret equivalent as:

$$A \equiv g^a \pmod{p}$$

$$B \equiv g^b \pmod{p}$$

$$A^b \equiv (g^a)^b \equiv g^{ab} \pmod{p}$$

$$B^a \equiv (g^b)^a \equiv g^{ba} \pmod{p}$$

Since $ab = ba$:

$$A^b \equiv B^a \pmod{p}$$

More explicitly:

$$(g^a)^b = (g^b)^a \pmod{p}$$

5.5. Worksheet Example (Diffie–Hellman)

Given: $p = 29$, $g = 2$, Alice secret $a = 11$, Bob secret $b = 7$.

Compute each step (leave your final value in the blank):

- Alice public value: $A \equiv g^a \pmod{29}$

$$A \equiv g^a \equiv 2^{11} \equiv \underline{\hspace{2cm}} \pmod{29}$$

So $A = \underline{\hspace{2cm}}$.

- Bob public value: $B \equiv g^b \pmod{29}$

$$B \equiv g^b \equiv 2^7 \equiv \underline{\hspace{2cm}} \pmod{29}$$

So $B = \underline{\hspace{2cm}}$.

- Shared secret (Alice): $s \equiv B^a \pmod{29}$

$$s \equiv B^a \equiv \underline{\hspace{2cm}} \pmod{29}$$

So $s = \underline{\hspace{2cm}}$.

- Shared secret (Bob): $s \equiv A^b \pmod{29}$

$$s \equiv A^b \equiv \underline{\hspace{2cm}} \pmod{29}$$

So $s = \underline{\hspace{2cm}}$.

Double-check that Alice's secret is equivalent to Bob's secret key!

6. RSA

RSA(Rivest-Shamir-Adleman) is a widely used public-key cryptosystem that relies on the difficulty of factoring large integers. It allows for secure communication and digital signatures.

6.1. Key Generation

A user generates an RSA keys pair as follows:

1. Choose two distinct large prime numbers p and q . (This are kept secret)
2. Compute $n = p * q$. (This is part of the public key)
3. Compute $\varphi(n) = (p - 1)(q - 1)$. (This is kept secret)
4. Choose an integer e such that $1 < e < \varphi(n)$ and $\gcd(e, \varphi(n)) = 1$. The integer e is the public exponent and is part of the public key. A common choice for e is $65537 = 2^{16} + 1$ to prevent small private exponents attack.
5. Compute private exponent d such that $d \equiv e^{-1} \pmod{\varphi(n)}$, d is the private exponent and should be kept secret.

To summarize, (n, e) is the public keys and (p, q, d) is the private keys.

6.2. Encryption

To encrypt a message M (where $0 \leq M < n$), the sender computes the ciphertext C using the recipient's public key (n, e) as follows:

$$C \equiv M^e \pmod{n}$$

6.3. Decryption

To decrypt the ciphertext C , the recipient uses their private key d to compute the original message M as follows:

$$M \equiv C^d \pmod{n}$$

This works because by Euler's theorem $a^{\varphi(n)} \equiv 1 \pmod{n}$:

$$C^d \equiv (M^e)^d \equiv M^{ed} \equiv M^{1+k\varphi(n)} \equiv M \pmod{n}$$

6.4. Worksheet Example (RSA)

Given: $p = 5$, $q = 11$, $e = 3$, message $M = 12$.

Fill in the blanks:

- $n = pq =$
- $\varphi(n) = (p-1)(q-1) =$
- Find $d \equiv e^{-1} : d =$ (mod $\varphi(n)$)
- Encrypt: $C \equiv M^e \equiv 12^3 =$ (mod n)
So $C =$
- Decrypt: $M' \equiv C^d \pmod{n} =$ (mod n)
So $M' =$

6.5. Security

The current best algorithm to attack RSA is to factor integers. Using the General Number Field Sieve, the time complexity is sub-exponential in the size of n or $f(x) = e^{(c+o(1))(\ln n)^{\frac{1}{3}}(\ln \ln n)^{\frac{2}{3}}}$, where $c = 1.923$.

Taking $o(1) = 20$ When $n = 2^{256}$, $f(n) \approx n^{2.08}$. At $n = 2^{2048}$, $f(n) \approx n^{0.651}$, roughly 10^{375} years of computation.

However, with Shor's algorithm, the time complexity is polynomial in the size of n , at roughly $O((\log n)^3)$. With 100k qubits, it is estimated that RSA-2048 can be broken in a few hours.

6.6. Wiener's attack

Theorem 6.6.1 (Wiener): Given a public key (e, n) where $n = pq$, if

$$d < \frac{1}{3}n^{\frac{1}{4}}$$

along with Legendre's theorem on continued fractions, where $|x - \frac{a}{b}| < \frac{1}{2b^2}$, then $\frac{a}{b}$ is one of the convergents fraction of $x - \frac{a}{b}$. Then d can be found using continued fractions quickly.

Proof:

Wiener's attack relies on the fact that if d is small, then the fraction $\frac{e}{n}$ can be approximated by a rational number with a small denominator, which corresponds to the private exponent d . This proof is challenging to follow, so do ask questions if needed.

6.6.1. Definition

To start off, let $G = \gcd(p, q)$ and let the Carmichael function be $\lambda(n) = \frac{\varphi(n)}{\gcd(p, q)} = \text{lcm}(p, q)$, then

$$ed = 1 + k\lambda(n)$$

6.6.2. Useful Lemmas

Lemma 1: $N - \varphi(n) < 3\sqrt{n}$

Proof: Assuming standard RSA encryptions, $p < q < 2q$. Using $N = pq$ and $p < \sqrt{n}$, we have $q < 2p = 2\sqrt{n}$ and thus $p + q - 1 < \sqrt{n} + 2\sqrt{n} - 1 = 2\sqrt{n} - 1 < 3\sqrt{n}$.

Since $\varphi(n) = (p-1)(q-1) = pq - p - q + 1$, $N - \varphi(n) = N - \varphi(n) = p + q - 1 < 3\sqrt{n}$

Lemma 2: $k < d$

Proof: $k\lambda(n) = ed - 1 < ed$. Since using standard RSA encryption, $e < \lambda(n)$, then $k\lambda(n) < ed < d\lambda(n)$. Cancelling $\lambda(n)$, we obtain $k < d$.

Lemma 3: $\frac{1}{2d} > \frac{1}{N^{\frac{1}{4}}}$

Proof: $d < \frac{1}{3}N^{\frac{1}{4}}$, $2d < 3d < N^{\frac{1}{4}}$, $\frac{1}{2d} > N^{\frac{1}{4}}$.

6.6.3. Proof

$$ed = 1 + k\lambda(n)$$

$$\frac{e}{\varphi(n)} - \frac{k}{Gd} = \frac{1}{d\varphi(n)}$$

Remember, our goal is to make an expression of $|x - \frac{a}{b}| < \frac{1}{2b^2}$ with known values. Thus, we approximate $\varphi(n)$ with n .

$$\left| \frac{e}{n} - \frac{k}{Gd} \right| = \left| \frac{edG - kN}{NGd} \right|$$

Since $k\varphi(n) = G(k\lambda(n)) = edG$ and starting from previous page:

$$\begin{aligned}
\left| \frac{edG - kN}{NGd} \right| &= \left| \frac{edG - k\varphi(n) - kN + k\varphi(n)}{NGd} \right| \\
&= \left| \frac{1 - k(N - \varphi(n))}{NGd} \right| \\
&< \left| \frac{1 - k(3\sqrt{n})}{NGd} \right| && \text{(Lemma 1)} \\
&< \left| \frac{3k\sqrt{n}}{NGd} \right| \\
&\leq \frac{3k}{d\sqrt{n}} \\
&< \frac{3d}{d\sqrt{n}} && \text{(Lemma 2)} \\
&= \frac{3\left(\frac{1}{3}N^{\frac{1}{4}}\right)}{d\sqrt{n}} \\
&= \frac{1}{dN^{\frac{1}{4}}} \\
&< \frac{1}{2d^2} && \text{(Lemma 3)}
\end{aligned}$$

Therefore,

$$\left| \frac{e}{n} - \frac{k}{Gd} \right| < \frac{1}{2d^2}$$

which satisfy the condition since $\frac{e}{n}$ is known.

Appendix

Supplemental Information

1. Bijection Proofs

1.1. Bijection Proof for Caesar or Affine Cipher

Theorem 1.1.1 (Affine):

Let $E(x) = ax + b \pmod{n}$ be the substitution function for the affine cipher, where n is prime so that for all a , $\gcd(a, n) = 1$. Then, E is a bijection from the set of integers modulo n to itself.

Proof:

Let $i, j \in \{1, 2, \dots, n-1\}$ be distinct integers, i.e. $i \neq j$. Without loss of generality (WLOG), let $i > j$. If $E(i) \equiv E(j) \pmod{n}$, we have:

$$ai + b \equiv aj + b \pmod{n}$$

$$a(i - j) \equiv 0 \pmod{n}$$

$$a(i - j) = kn$$

Since $\gcd(a, n) = 1$, by euclid lemma, $n \mid (i - j)$. However, since $0 < i - j < n$, $i - j$ is not divisible by n . This is a contradiction and thus $i = j$.

Via pigeonhole principle, since there are n possible outputs and n possible inputs, and no two input map to the same output, each input must map to a unique output. Hence, E is a bijection.

This also proves caesar cipher is a bijection as it is a special case of affine cipher where $a = 1$.

2. Proof of Fermat's Little Theorem (FLT)

We can proof via more generalised Euler's theorem $a^{\varphi(n)} \equiv 1 \pmod{n}$ or formal proof of FLT using binomial distribution. We will choose the latter, which is more intuitive and easier to understand. (To prove Euler's theorem, it's further down in the appendix)

Theorem 2.1 (FLT):

If p is a prime and a is an integer not divisible by p , then

$$a^{p-1} \equiv 1 \pmod{p}$$

Or similarly:

$$a^p \equiv a \pmod{p}$$

Proof:

Consider the expansion of $(1 + a)^p$ using the binomial theorem:

Let p be a prime number and a an integer.

We start with the binomial expansion:

$$(1 + a)^p = \sum_{k=0}^p \binom{p}{k} 1^{p-k} a^k = \sum_{k=0}^p \binom{p}{k} a^k$$

Expanding the sum, we have:

$$(1 + a)^p = \binom{p}{0} a^0 + \binom{p}{1} a^1 + \binom{p}{2} a^2 + \dots + \binom{p}{p-1} a^{p-1} + \binom{p}{p} a^p$$

Since:

$$\binom{p}{0} = 1, \quad \binom{p}{p} = 1$$

So:

$$(1 + a)^p = 1 + a^p + \sum_{k=1}^{p-1} \binom{p}{k} a^k$$

Now observe that for $1 \leq k \leq p-1$:

$$\binom{p}{k} = \frac{p!}{k!(p-k)!}$$

Since p is prime, p divides $p!$ but does not divide $k!$ or $(p-k)!$. Hence:

$$p \mid \binom{p}{k}$$

Therefore, for all $1 \leq k \leq p-1$:

$$\binom{p}{k} \equiv 0 \pmod{p}$$

So all middle terms vanish modulo p :

$$\sum_{k=1}^{p-1} \binom{p}{k} a^k \equiv 0 \pmod{p}$$

Thus, the inductive step is shown:

$$(a + 1)^p \equiv a^p + 1 \pmod{p}$$

Via induction, $1^p \equiv (0^p + 1) \pmod{p} \Rightarrow a = 0$ is true and given the inductive step:

$$a^p \equiv a \pmod{p}$$

By the principle of mathematical induction, we have proved Fermat's Little Theorem.

3. Euler's Theorem

Theorem 3.1 (Euler):

If $a, n \in \mathbb{Z}$ such that $\gcd(a, n) = 1$. Then:

$$a^{\varphi(n)} \equiv 1 \pmod{n}$$

Proof: Consider the set of integers $S = \{r_1, r_2, \dots, r_{\varphi(n)}\}$ which represent the reduced residue system modulo n . These are the integers in the range $[1, n]$ that are coprime to n .

Since $\gcd(a, n) = 1$, we can multiply each element in S by a to form a new set:

$$S' = \{ar_1, ar_2, \dots, ar_{\varphi(n)}\}$$

We claim that S' is also a reduced residue system modulo n , which is similar to bijection proof for affine cipher. Therefore, the set S' is simply a permutation of S modulo n . Multiplying the elements of both sets together gives:

$$\prod_{i=1}^{\varphi(n)} ar_i \equiv \prod_{i=1}^{\varphi(n)} r_i \pmod{n}$$

Factoring out a from the product on the left:

$$a^{\varphi(n)} (r_1 r_2 \dots r_{\varphi(n)}) \equiv (r_1 r_2 \dots r_{\varphi(n)}) \pmod{n}$$

Let $P = \prod_{i=1}^{\varphi(n)} r_i$. Since each r_i is coprime to n , their product P is also coprime to n . This means P has a multiplicative inverse modulo n . Multiplying both sides by P^{-1} yields:

$$a^{\varphi(n)} \equiv 1 \pmod{n}$$